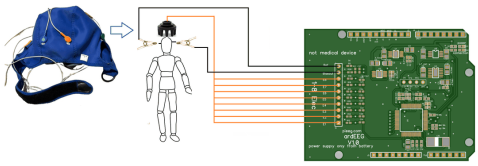
CONNECTOR PINOUT & ELECTRODE GUIDE · SKU CQA240229AR

Electrodes, Header & Arduino Pins

How to connect 8 electrodes, the reference and the bias to ardEEG, a suggested 10-20 montage, the five Arduino pins the shield uses, and the LEDs and test points on the board.



8	REF	BIAS	5	2.54
Electrode channels	Ear clip · common 0 µV	Ear clip · noise rejection	Arduino signal pins	mm header pitch

01 ELECTRODE INPUTS

Input	Count	Electrode	Purpose
Channels 1–8	8	Dry or wet Ag/AgCl, in the cap	Voltage between each electrode and the reference
Reference	1	Ear clip on one earlobe	Common 0 µV point for all 8 channels
Bias	1	Ear clip on the other earlobe	Drives the body to cancel common-mode noise such as 50/60 Hz hum

02 ELECTRODE HEADER J3

Square pad = pin 1 (E1). 2.54 mm pitch, single row.

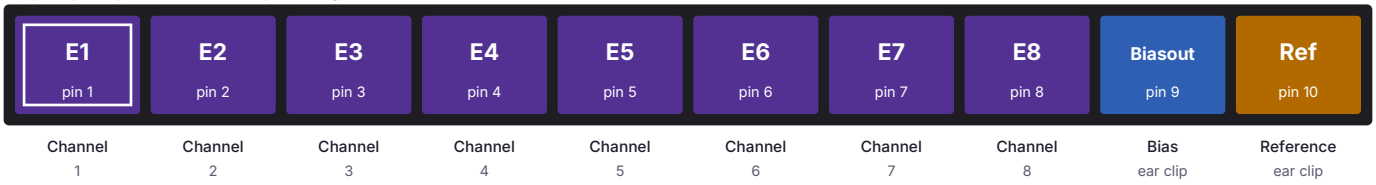


Figure 1 — Electrode header pin order. On the board E1 is nearest the bottom edge and Ref at the top; silkscreen “1-8 Elec”.

NOTE Reference and bias clips matter most.

A loose ear clip makes every channel noisy at once. Clean the earlobes and make sure both clips sit firmly. Use 2.54 mm (0.1") connectors; the header is shrouded, so plug in straight.

03 LEDS AND TEST POINTS

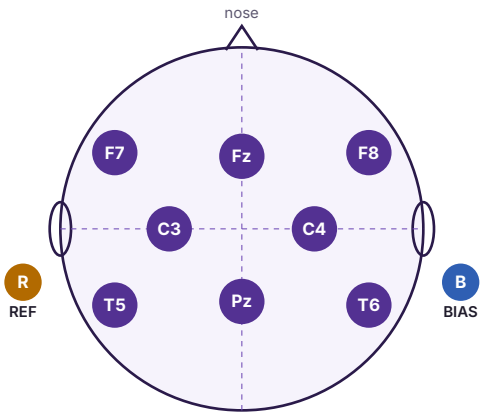
LED	Meaning
D1 “power”	Shield powered
D2 “connection”	ADS1299 connected / running

J1 pin	Expected	Use
−2.5V	≈ −2.5 V	Supply check
2.5V	≈ +2.5 V	Supply check
3.2V	≈ 3.2 V	Supply check
gnd	0 V	Meter reference

Measure J1 with a multimeter only, with no electrodes on anyone. Do not draw current from the test points.

For research and education only. Not a medical device. Specifications subject to change.

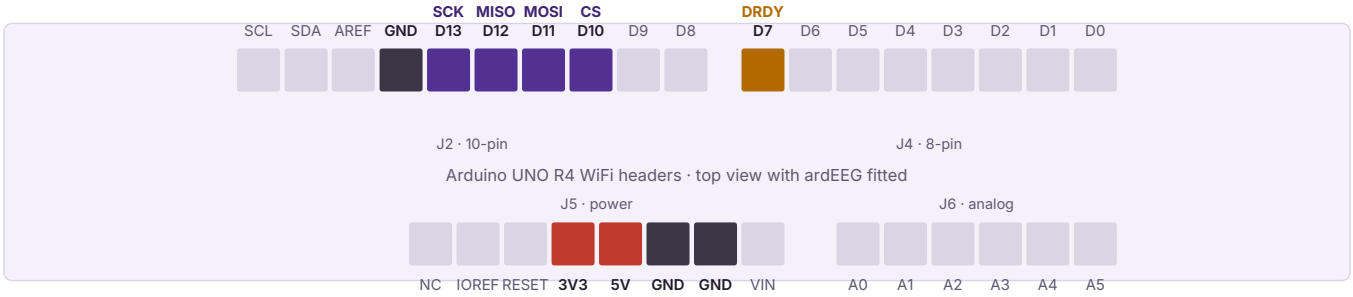
04 SUGGESTED MONTAGE (10-20 SYSTEM)



Position	Area	Good for
F7, F8	Front left / right, side	Eye movement, frontal activity
Fz	Front centre	Blinks, attention; example recordings
C3, C4	Centre left / right	Motor activity (hands)
T5, T6	Back left / right, side	Temporal activity
Pz	Back centre, top	Alpha rhythm

Assign the positions to channels 1–8 in any order and note which went where — the software only knows channel numbers.

05 ARDUINO PINS



Arduino pin	Signal	Direction
D13	SPI clock (SCK)	Arduino → ADS1299
D12	SPI MISO (DOUT)	ADS1299 → Arduino
D11	SPI MOSI (DIN)	Arduino → ADS1299
D10	Chip select (CS)	Arduino → ADS1299
D7	DRDY — data ready, active low, 250 per second	ADS1299 → Arduino

The shield also takes 5 V, 3.3 V and GND from the power header. All other pins — D0–D6, D8, D9, A0–A5, SCL/SDA — stay free for your own sensors, buttons, LEDs or motors.

06 RELATED DOCUMENTS

AE-DS-001 Product Datasheet	**AE-MD-002** Mechanical & Dimensional Drawing	**AE-PCB-003** PCB & Manufacturing Specification
AE-QSG-005 Quick Start & Safety Guide	**AE-SW-006** Data Interface & Software Guide	

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